

Example 1 | data/verified_kinhub.jsonl line 8301

correction_type	Km value correction: source row 0.055 -> paper-grounded 55 mM.
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top-level fields

source_db	brenda
ec_number	1.1.1.61
organism	Clostridium kluyveri
enzyme_name	4-hydroxybutyrate dehydrogenase
uniprot_id	P38945
kcat_unit	s ⁽⁻¹⁾
km	55.0
km_unit	mM
kcat_over_km_unit	s ⁽⁻¹⁾ *mM ⁽⁻¹⁾
supporting_pmids	["7606170"]
supporting_dois	["10.1006/prep.1995.1026"]
verification_status	corrected
verification_method	paper_grounding_verification
evidence_confidence_tier	paper_grounding
paper_grounding_status	review_excerpt_preserved

correction/evidence fields

review_summary	Corrected the Clostridium kluyveri 4-hydroxybutanoate Km row to the paper-grounded value of 55 mM; the current 0.055 value is a unit-scale mismatch against PMID 7606170.
critique_rationale	A verified outcome would preserve the current numeric claim, but the abstract directly contradicts that value. The same PMID supports keeping the row in export only after correcting the Km to 55 mM and preserving the cited evidence, so corrected is the defensible outcome.
agentic_understanding	The identity side of this record is already coherent across BRENDA and UniProt: Clostridium kluyveri, UniProt P38945, EC 1.1.1.61, and 4-hydroxybutanoate as the reverse-direction substrate for the same oxygen-sensitive dehydrogenase paper. The paper-grounding pass shows that the row is not merely missing evidence; one structured source currently carries the wrong scale because PMID 7606170 reports the 4-hydroxybutanoate Km in the reverse direction as 55 mM, while the pending row still holds 0.055.
verification_notes	PMID: 7606170 DOI: 10.1006/prep.1995.1026 Quote: "In the reverse direction, the Km values were 670 +/- 80 microM and 55 +/- 16 mM for NAD and 4-hydroxybutanoate, respectively." Evidence: This directly grounds the Clostridium kluyveri 4-hydroxybutanoate row and shows the current 0.055 entry is off by three orders of magnitude; the paper-backed correction is 55 mM.
audit_history	["Loop 1: Confirmed the batch identity snapshot is consistent for Clostridium kluyveri, UniProt P38945, and the 4-hydroxybutanoate/NAD couple anchored to PMID 7606170.", "Loop 2: Fetched the PMID 7606170 abstract and extracted the reverse-direction kinetic sentence covering NAD and 4-hydroxybutanoate.", "Loop 3: Finalized the row as corrected because the abstract reports 4-hydroxybutanoate Km as 55 +/- 16 mM, not 0.055 mM."]

raw_data field

raw_data	<pre> {"abstract_identity_anchor_hits": [], "anomaly_reasons": [], "evidence_confidence_note": "This record preserves review-authored paper evidence excerpts with explicit citation context.", "evidence_confidence_tier": "paper_grounded", "evidence_envelope": {"anomaly": {"reasons": [], "source_record_count": 2}, "identity": {"ec_number": "1.1.1.61", "max_sequence_source_confidence": 1.0, "mutation_evidence_notes": [], "mutation_evidence_sources": [], "mutation_key": "unknown_state", "mutation_signatures": [], "mutation_types": [], "organisms": ["Clostridium kluyveri", "Clostridium kluyveri (strain ATCC 8527 / DSM 555 / NBRC 12016 / NCIMB 10680 / K1)", "sequence": "MKLLKLAPDVYK FDTAEFMKYFKVGKGFILTFNEFLYKPFLEKFNDGADAVFQEKYGLGEPSEDEMNNIIKDIGDKQYNRIIIVGGGSVIDIAKI LSLKYTDDSLDLFEGKVPLVKNKELIIVPTTCGTGSEVTNVSVaelKRRHTKKGIASDELATYAVLVPEFIKGLPYKFFVTSS VDALIHATEAYVSPNANPYTDMFSVKAMELILNGYMQMVEKGNDYRVEIEDFVIGSNYAGIAFGNAGVGAVHALSYPIGGN YHVPHGAEANYLFFTEIFKTYYEKNPNNGKIKDVNKLKLAGILKCDESEAYDSLSQLLDKLLSRKPLREYGMKEEEIETFADSVIE GQQRLLVNNYEPFSREDIVNTYKKLY", "sequence_resolved": true, "sequence_sources": ["uniprot_ec_organism_unique", "source_record"], "smiles": "C(CC(=O)[O-])CO", "smiles_values": ["C(CC(=O)[O-])CO"], "substrate_name_candidates": ["4-hydroxybutanoate"], "uniprot_ids": ["P38945"]}, {"cross_source_identity_matches": {"shared_organisms": ["Clostridium kluyveri", "Clostridium kluyveri (strain ATCC 8527 / DSM 555 / NBRC 12016 / NCIMB 10680 / K1)", "shared_smiles_values": ["C(CC(=O)[O-])CO"], "shared_substrate_name_candidates": ["4-hydroxybutanoate"], "shared_uniprot_ids": ["P38945"], "sources_with_smiles": ["brenda", "uniprot"], "sources_with_uniprot_id": ["brenda", "uniprot"]}, {"source_identity_hints": [{"organism": "Clostridium kluyveri", "pubmed_id": "7606170", "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "smiles": "C(CC(=O)[O-])CO", "source_db": "brenda", "substrate_name_candidates": ["4-hydroxybutanoate"], "uniprot_id": "P38945"}, {"organism": "Clostridium kluyveri (strain ATCC 8527 / DSM 555 / NBRC 12016 / NCIMB 10680 / K1)", "pubmed_id": "7606170", "sequence_source": "source_record", "sequence_source_confidence": 1.0, "smiles": "C(CC(=O)[O-])CO", "source_db": "uniprot", "substrate_name_candidates": ["4-hydroxybutanoate"], "uniprot_id": "P38945"]}], "provenance": {"paper_mentions": [], "source_databases_merged": ["brenda", "uniprot"], "supporting_dois": [], "supporting_pmids": ["7606170"]}, "schema_version": 1, "source_records": [{"km_unit": "mM", "km_value": 0.055, "organism": "Clostridium kluyveri", "pubmed_id": "7606170", "raw_data": {"brenda_reference_id": "287171", "brenda_reference_metadata": {"authors": "Wolff, R.A.; Kenealy W.R.", "journal": "Protein Expr. Purif.", "pages": "206-212", "pubmed_id": "7606170", "title": "Purification and characterization of the oxygen-sensitive 4-hydroxybutanoate dehydrogenase from Clostridium kluyveri", "volume": "6", "year": "1995"}, "ecNumber": "1.1.1.61", "kmValue": "0.055", "ligandStructureId": 2192, "literature": [287171], "organism": "Clostridium kluyveri", "sequence_enrichment": {"sequence": " MKLLKLAPDVYKFDTAEEFMKYFKVGKGFILTFNEFLYKPFLEKFNDGADAVFQEKYGLGEPSEDEMNNIIKDIGDKQYNRII AVGGGSVIDIAKILSLKYTDDSLDLFEGKVPLVKNKELIIVPTTCGTGSEVTNVSVaelKRRHTKKGIASDELATYAVLVPEFI KGLPYKFFVTSSVDALIHATEAYVSPNANPYTDMFSVKAMELILNGYMQMVEKGNDYRVEIEDFVIGSNYAGIAFGNAGVG AVHALSYPIGGNYHVPHGAEANYLFFTEIFKTYYEKNPNNGKIKDVNKLKLAGILKCDESEAYDSLSQLLDKLLSRKPLREYGMK EEEIETFADSVIEGQQRLLVNNYEPFSREDIVNTYKKLY", "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "uniprot_id": "P38945"}, "sequence_variant": {"sequence_variant_note": "Sequence identity was resolved, but no explicit wild-type or mutant call was captured.", "sequence_variant_status": "sequence_identity_only"}, "smiles_enrichment": {"matched_name": "4-hydroxybutanoate", "original_substrate": "4-hydroxybutanoate", "smiles": "C(CC(=O)[O-])CO", "substrate": "4-hydroxybutanoate"}, "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "smiles": "C(CC(=O)[O-])CO", "source_db": "brenda", "substrate": "4-hydroxybutanoate", "substrate_name_candidates": ["4-hydroxybutanoate"], "uniprot_id": "P38945"}, {"km_unit": "mM", "km_value": 55.0, "organism": "Clostridium kluyveri (strain ATCC 8527 / DSM 555 / NBRC 12016 / NCIMB 10680 / K1)", "pubmed_id": "7606170", "raw_data": {"km_entry": {"constant": 55.0, "evidences": [{"evidenceCode": "ECO:0000269", "id": "7606170", "source": "PubMed"}], "substrate": "4-hydroxybutanoate", "unit": "mM"}, "note_context": "", "sequence_variant": {"sequence_variant_note": "Sequence identity was resolved, but no explicit wild-type or mutant call was captured.", "sequence_variant_status": "sequence_identity_only"}, "smiles_enrichment": {"matched_name": "4-hydroxybutanoate", "original_substrate": "4-hydroxybutanoate", "smiles": "C(CC(=O)[O-])CO", "uniprot_accession": "P38945"}, "sequence_source": "source_record", "sequence_source_confidence": 1.0, "smiles": "C(CC(=O)[O-])CO", "source_db": "uniprot", "substrate": "4-hydroxybutanoate", "substrate_name_candidates": ["4-hydroxybutanoate"], "uniprot_id": "P38945"}], "evidence_schema_version": 1, "paper_grounding_note": "Verification notes preserve a cited paper excerpt from manual review.", "paper_grounding_source": "review_notes_excerpt", "paper_grounding_status": "review_excerpt_preserved", "paper_mentions": [], "paper_structure_signals": [], "sequence_resolved": true, "source_ ... [raw_data truncated in PDF; full row in JSONL] </pre>
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Example 2 | data/verified_kinhub.jsonl line 11931

correction_type	kcat value correction: extracted kcat conflicts with full-text Table 1.
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top-level fields

source_db	brenda
ec_number	1.1.1.149
organism	Oryctolagus cuniculus
uniprot_id	P80508
kcat	0.153
kcat_unit	s ⁽⁻¹⁾
km	0.0034
km_unit	mM
kcat_over_km	45.0
kcat_over_km_unit	s ⁽⁻¹⁾ *mM ⁽⁻¹⁾
supporting_pmids	["23995665"]
verification_status	corrected
verification_method	full_text_table_review
evidence_confidence_tier	paper_grounded_high_confidence
paper_grounding_status	full_text_verified

correction/evidence fields

review_summary	Full-text Table 1 supports the substrate identity, but the extracted kcat does not match the paper, so the row is corrected.
critique_rationale	The paper is explicit about 5alpha-dihydrotestosterone and its kcat, but the source extraction understates that rate. The preserved note must therefore record a corrected paper-grounded verdict rather than a plain verification.
agentic_understanding	anchored the row to PMID 23995665 and Table 1. checked the published Km/kcat pair against the extracted BRENDA row. preserved the full-text table excerpt and finalized the corrected verdict.
verification_notes	[FULL TEXT KINETIC EXTRACT] PMID: 23995665 Quote: "5alpha-Dihydrotestosterone 18 2.8 0.16" Evidence: The published kcat is 2.8 min ⁻¹ , which conflicts with the extracted 0.025 s ⁻¹ , so the row is corrected to the paper-backed value.
audit_history	["Loop 1: anchored PMID 23995665 and the Table 1 entry for 5alpha-dihydrotestosterone.", "Loop 2: compared the published Km/kcat pair against the extracted BRENDA row.", "Loop 3: preserved the full-text excerpt and marked the row corrected because the extracted kcat does not match the paper exactly."]

raw_data field

Example 3 | data/verified_kinhub.jsonl line 19409

correction_type	Km value correction: source row 0.003 mM -> paper table 3 mM.
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top-level fields

source_db	brenda
ec_number	1.1.3.38
organism	Penicillium simplicissimum
uniprot_id	P56216
kcat	4.2
kcat_unit	s ⁻¹
km	0.003
km_unit	mM
kcat_over_km	1400.0
kcat_over_km_unit	s ⁻¹ *mM ⁻¹
supporting_pmids	["10099328", "9791114", "9654070"]
verification_status	corrected
verification_method	paper_grounded_verification_review
evidence_confidence_tier	paper_grounded
paper_grounding_status	review_excerpt_preserved

correction/evidence fields

review_summary	Correct the 4-propylphenol Km to 3 mM from PMID 10099328.
critique_rationale	A conservative unverified verdict would leave a known 1000x unit/extraction mismatch in the export. The lead paper directly quotes the correct Km value, so correcting Km to 3 mM is less speculative than preserving the current merged value.
agentic_understanding	The batch is a BRENDA merge for vanillyl alcohol oxidase from Penicillium simplicissimum. The lead paper for PMID 10099328 directly reports the exact 4-propylphenol table row, and that paper matches the enzyme/substrate context plus a Km of 3 mM. Related propylphenol papers in the same batch support the broader literature context, but they do not justify keeping the current 0.003 mM Km when the lead paper gives the explicit 3 mM value.
verification_notes	PMID: 10099328 Quote: "4-Propylphenol 3.9 3" Evidence: Table I directly reports the VAO / Penicillium simplicissimum 4-propylphenol row with kcat 3.9 s ⁻¹ and Km 3 mM, so the batch Km should be corrected from 0.003 mM to 3 mM. Related propylphenol papers in the batch provide broader literature context, but this lead-paper table is the exact match for the corrected row.
audit_history	["Loop 1: anchored the row to the identity snapshot and confirmed the batch is a paper-grounded BRENDA merge around Penicillium simplicissimum VAO.", "Loop 2: fetched the lead PMID 10099328 table and cross-checked the related propylphenol papers for context.", "Loop 3: finalized a correction because the lead paper supports Km = 3 mM, not the batch's current 0.003 mM."]]

raw_data field

raw_data	<pre> {"abstract_identity_anchor_hits": [], "anomaly_reasons": [], "evidence_confidence_note": "This record preserves review-authored paper evidence excerpts with explicit citation context.", "evidence_confidence_tier": "paper_grounded", "evidence_envelope": {"anomaly": {"reasons": [], "source_record_count": 6}, "identity": {"ec_number": "1.1.3.38", "max_sequence_source_confidence": 0.6, "mutation_evidence_notes": [], "mutation_evidence_sources": [], "mutation_key": "unknown_state", "mutation_signatures": [], "mutation_types": [], "organisms": ["Penicillium simplicissimum"], "sequence": "MSKTQEFRPLTLPPKLSLSDNEFIQDIIRIVGSENVVSSKDQIVDGSYMKPHTHTDPPHH VMDQDYFLASAI VAPRNVADVQSIVGLANKFSFPLWPISIGRNSGYGGAAPRVSGSVVLDMGKNMNRVLEVNVEGAYCVV EPGVTYHDLHNYLEANNLRDKLWLDVDPDLGGGSVLGNAVERGVGYTPYGDHWMHSGMEVVLANGELLRTGMGALPD PKRPETMGLKPEDQPWSKIAHLFPYGFPGPYIDGLFSQSNMGIVTKIGIWLMPNPRGYQSYLITLPKDGDLKQAVDIIRPLRL GMALQNVPTIRHILLDAAVLGDKRSYSSRTEPLSDEELDKIAKQLNLGRWNFYGALYGPEPIRRVLWETIKDAFSAIPGVKF YFPEDTPENSVL RVRDKTMQGIPTYDELKWDWLPNGAHLFFSPIAKVSGEDAMMQYAVTKKRCQEAGLDFIGTFTVGM EMHHIVCIVFNKDLIQKRKVQWLMRTLIDDCANGWGEYRTHLAFMDQIMETYNWNNSSFLRFNEVLKNAVDPNPNIAPG KSGVWPSQYSHVTWKL", "sequence_resolved": true, "sequence_sources": ["uniprot_ec_organism_unique"], "smiles": "CCCC1=CC=C(C=C1)O", "smiles_values": ["CCCC1=CC=C(C=C1)O"], "substrate_name_candidates": ["4-Propylphenol", "4-n-Propylphenol"], "uniprot_ids": ["P56216"]}, {"identity_hints": {"cross_source_identity_matches": {"shared_organisms": ["Penicillium simplicissimum"], "shared_smiles_values": ["CCCC1=CC=C(C=C1)O"], "shared_substrate_name_candidates": ["4-Propylphenol", "4-n-Propylphenol"], "shared_uniprot_ids": ["P56216"], "sources_with_smiles": ["brenda"], "sources_with_uniprot_id": ["brenda"]}, {"organism": "Penicillium simplicissimum", "pubmed_id": "10099328", "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "smiles": "CCCC1=CC=C(C=C1)O", "source_db": "brenda", "substrate_name_candidates": ["4-Propylphenol"], "uniprot_id": "P56216"}, {"organism": "Penicillium simplicissimum", "pubmed_id": "9791114", "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "smiles": "CCCC1=CC=C(C=C1)O", "source_db": "brenda", "substrate_name_candidates": ["4-n-Propylphenol"], "uniprot_id": "P56216"}, {"organism": "Penicillium simplicissimum", "pubmed_id": "9654070", "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "smiles": "CCCC1=CC=C(C=C1)O", "source_db": "brenda", "substrate_name_candidates": ["4-n-Propylphenol"], "uniprot_id": "P56216"}, {"organism": "Penicillium simplicissimum", "pubmed_id": "10099328", "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "smiles": "CCCC1=CC=C(C=C1)O", "source_db": "brenda", "substrate_name_candidates": ["4-Propylphenol"], "uniprot_id": "P56216"}, {"organism": "Penicillium simplicissimum", "pubmed_id": "9654070", "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "smiles": "CCCC1=CC=C(C=C1)O", "source_db": "brenda", "substrate_name_candidates": ["4-Propylphenol"], "uniprot_id": "P56216"}, {"organism": "Penicillium simplicissimum", "pubmed_id": "9791114", "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "smiles": "CCCC1=CC=C(C=C1)O", "source_db": "brenda", "substrate_name_candidates": ["4-Propylphenol"], "uniprot_id": "P56216"}]}, {"provenance": {"paper_mentions": [], "source_databases_merged": ["brenda"], "supporting_dois": [], "supporting_pmids": ["10099328", "9791114", "9654070"]}, "schema_version": 1, "source_records": [{"kcat_unit": "s^-1", "kcat_value": 3.9, "organism": "Penicillium simplicissimum", "pubmed_id": "10099328", "raw_data": {"brenda_reference_id": "10671", "brenda_reference_metadata": {"authors": "Drijfhout, F.P.; Fraaije, M.W.; Jongejan, H.; van Berkel, W.J.H.; Franssen, M.C.R.", "journal": "Biotechnol. Bioeng.", "pages": "171-177", "pubmed_id": "10099328", "title": "Enantioselective hydroxylation of 4-alkylphenols by vanillyl alcohol oxidase", "volume": "59", "year": "1998"}, "ecNumber": "1.1.3.38", "ligandStructureId": 6094, "literature": [10671], "organism": "Penicillium simplicissimum", "sequence_enrichment": {"query_organism": "Penicillium simplicissimum", "sequence": "MSKTQEFRPLTLPPKLSLSDNEFIQDIIRIVGSENVVSSKDQIVDGSYMKPHTHTDPPHHVMDQDYFLASAI VAPRNVADVQSIVGLANKFSFPLWPISIGRNSGYGGAAPRVSGSVVLDMGKNMNRVLEVNVEGAYCVV EPGVTYHDLHN YLEANNLRDKLWLDVDPDLGGGSVLGNAVERGVGYTPYGDHWMHSGMEVVLANGELLRTGMGALPD PKRPETMGLK PEDQPWSKIAHLFPYGFPGPYIDGLFSQSNMGIVTKIGIWLMPNPRGYQSYLITLPKDGDLKQAVDIIRPLRLGMALQNVPTIRH ILLDAAVLGDKRSYSSRTEPLSDEELDKIAKQLNLGRWNFYGALYGPEPIRRVLWETIKDAFSAIPGVKFYFPEDTPENSVL RVRDKTMQGIPTYDELKWDWLPNGAHLFFSPIAKVSGEDAMMQYAVTKKRCQEAGLDFIGTFTVGMREMHHIVCIVFNK DLIQKRKVQWLMRTLIDDCANGWGEYRTHLAFMDQIMETYNWNNSSFLRFNEVLKNAVDPNPNIAPGKSGVWPSQYS HVTWKL", "sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "uniprot_id": "P56216"}, "sequence_variant": {"sequence_variant_note": "Sequence identity was resolved, but no explicit wild-type or mutant call was captured.", "sequence_variant_status": "sequence_identity_only"}, "smiles_enrichment": {"matched_name": "4-Propylphenol", "original_substrate": "4-Propylphenol", "smiles": "CCCC1=CC=C(C=C1)O"}, "substrate": "4-Propylphenol", "turnoverNumber": "3.9"}, {"sequence_source": "uniprot_ec_organism_unique", "sequence_source_confidence": 0.6, "smiles": "CCCC1=CC=C(C=C1)O", "source_db": "... [raw_data truncated in PDF; full row in JSONL] </pre>
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Example 4 | data/verified_kinhub.jsonl line 8538

correction_type	Identity/EC correction: wrong AKR1C / EC branch -> HSD3B1 / EC 1.1.1.145.
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top-level fields	
source_db	brenda
ec_number	1.1.1.145
organism	Homo sapiens
kcat_unit	s ⁻¹
km	1e-05
km_unit	mM
kcat_over_km_unit	s ⁻¹ *mM ⁻¹
supporting_pmids	["9287397", "1401999"]
supporting_dois	["10.1159/000211481"]
verification_status	corrected
verification_method	batch_brenda_ambiguous_ec_organism__homo_sapiens__9287397__part01_review
evidence_confidence_tier	paper_grounded
paper_grounding_status	review_excerpt_preserved

correction/evidence fields	
review_summary	Corrected the record from AKR1C/EC 1.1.1.51 ambiguity to human skin 3-beta-hydroxysteroid dehydrogenase type 1 (HSD3B1, UniProt P14060) based on the cited PMID 9287397 assay context and reviewed UniProt skin expression evidence.
critique_rationale	Keeping the record in manual review would preserve a false ambiguity created by the wrong EC branch. The decisive contradiction is that PMID 9287397 explicitly studies human skin 3-beta-hydroxysteroid dehydrogenase/delta-5-4-isomerase with dehydroepiandrosterone substrate, while the batch candidate pool contains AKR1C family entries for EC 1.1.1.51 and no evidence linking them to the cited assay. Between the two reviewed human 3-beta-HSD isoforms, UniProt tissue specificity provides the needed discriminator: HSD3B1 is annotated for skin and HSD3B2 is not. I chose corrected rather than verified because the main repair is identity correction from an incorrect EC/accession family onto the reviewed human skin isoform.
agentic_understanding	anchored on the identity_resolution_snapshot, which showed that the saved row had no preserved sequence or accession and had been routed into an AKR1C-family EC 1.1.1.51 candidate pool solely from ambiguous EC+organism lookup. checked the cited paper directly and found that PMID 9287397 is not an AKR1C paper at all: it reports activity of 3-beta-hydroxysteroid dehydrogenase/delta-5-4-isomerase in human skin using dehydroepiandrosterone as substrate and reports formation of 4-androstene-3,17-dione, which matches EC 1.1.1.145 plus steroid isomerase activity rather than 3-alpha-hydroxysteroid dehydrogenase. reviewed human UniProt entries for this enzyme class and found that P14060 (HSD3B1) is the reviewed human type 1 3-beta-HSD entry explicitly annotated for skin, whereas P26439 (HSD3B2) is adrenal/gonadal. That tissue-specific discriminator resolves the identity to HSD3B1 and supports a corrected sequence-backed verdict.
verification_notes	Context: the selected BRENDA row was stored as EC=1.1.1.51, organism=Homo sapiens, substrate=dehydroepiandrosterone, but had no preserved sequence/accession and only an ambiguous EC+organism lookup candidate pool. Evidence used: PMID: 9287397; DOI: 10.1159/000211481. Quote: "Activity and inhibition of 3-beta-hydroxysteroid dehydrogenase/delta 5-4-isomerase in human skin." Quote: "The Michaelis-Menten constant of the enzyme was found to be Km = 10nM and the maximal velocity was Vmax = 0.625 pmol produced 4-androstene-3,17-dione/mg protein/20 min." Reviewed UniProt follow-up showed P14060 = human 3 beta-hydroxysteroid dehydrogenase/Delta 5-->4-isomerase type 1 with tissue specificity "Placenta and skin (PubMed:1401999)", whereas P26439 = type 2 is expressed in adrenal gland, testis and ovary. Resolution: PMID 9287397 supports human skin 3-beta-HSD activity, not the original AKR1C/EC 1.1.1.51 branch; the skin tissue context resolves the reviewed human isoform to HSD3B1 / P14060. Decision: corrected to EC 1.1.1.145 with sequence_source>manual_pmid_uniprot_reviewed_match and UniProt P14060 because the cited assay context plus reviewed tissue specificity break the original wrong-family ambiguity.
audit_history	["Loop 1: Read the identity_resolution_snapshot first and confirmed the row had no preserved sequence or UniProt id, one BRENDA source record, dehydroepiandrosterone substrate, Homo sapiens organism, PMID 9287397, and a four-way zero-score AKR1C candidate pool under EC 1.1.1.51.", "Loop 2: Queried PMID 9287397 directly and found the abstract/title describe \"Activity and inhibition of 3-beta-hydroxysteroid dehydrogenase/delta-5-4-isomerase in human skin\" with dehydroepiandrosterone substrate and product formation of 4-androstene-3,17-dione, contradicting the original AKR1C/EC 1.1.1.51 assignment.", "Loop 3: Reviewed human UniProt 3-beta-HSD entries and finalized the correction because P14060 (HSD3B1) is the reviewed human type 1 enzyme annotated for skin, while P26439 (HSD3B2) is annotated for adrenal gland, testis, and ovary rather than skin."]

raw_data field

raw_data	<pre> {"abstract_identity_anchor_hits": [], "anomaly_reasons": [], "evidence_confidence_note": "This record preserves review-authored paper evidence excerpts with explicit citation context.", "evidence_confidence_tier": "paper_grounded", "evidence_envelope": {"anomaly": {"reasons": [], "source_record_count": 1}, "identity": {"ec_number": "1.1.1.51", "mutation_evidence_notes": [], "mutation_evidence_sources": [], "mutation_key": "unknown_state", "mutation_signatures": [], "organisms": ["Homo sapiens"], "sequence": "MTGWSCLVTGAGGFLGQRIIRLLVKEKELKEIRVLDKAFGPRLREEFSKL QNKTCLTVLEGDILDEPFLKRACQDVSIIHTACIIDVFGVTHRESIMNVNVKGTQLLLEACVQASVPVFIYTSSIEVAGPNSY KEIIQNGHEEEPLENTWPAPYPHSHKLAEKAVLAANGWNLKNGGTLTYTCALRPMYIYEGESRFLSASINEALNNNGILSSV GKFSTVNPVYGVNVAWAHILRALQDPKAPSIRGQFYISDDTPHQSYDNLNYTSLKEFGLRLDSRWSFPLSLMYWIGF LLEIVSFLLRPIYTYRPPFNRHIVTLNSVFTFSYKKAQRDLAYKPLYSWEEAKQKTVEWVGSVLDRHKETLKSQTQ", "sequence_resolved": true, "sequence_sources": [], "smiles": "CC12CCC3C(C1CCC2=O)CC=C4C3(CCC(C4)O)C", "smiles_values": ["CC12CCC3C(C1CCC2=O)CC=C4C3(CCC(C4)O)C"]}, "substrate_name_candidates": ["dehydroepiandrosterone"], "uniprot_ids": [], "identity_hints": {"cross_source_identity_matches": {"shared_organisms": ["Homo sapiens"], "shared_smiles_values": ["CC12CCC3C(C1CCC2=O)CC=C4C3(CCC(C4)O)C"], "shared_substrate_name_candidates": ["dehydroepiandrosterone"], "shared_uniprot_ids": [], "sources_with_smiles": ["brenda"], "sources_with_uniprot_id": [], "source_identity_hints": [{"organism": "Homo sapiens", "pubmed_id": "9287397", "smiles": "CC12CCC3C(C1CCC2=O)CC=C4C3(CCC(C4)O)C", "source_db": "brenda", "substrate_name_candidates": ["dehydroepiandrosterone"]}]}, "provenance": {"paper_mentions": ["37°C, pH not specified in the publication"], "source_databases_merged": "brenda", "supporting_dois": [], "supporting_pmids": ["9287397"]}, "schema_version": 1, "source_records": [{"km_unit": "mM", "km_value": 1e-05, "organism": "Homo sapiens", "paper_note": "37°C, pH not specified in the publication", "pubmed_id": "9287397", "raw_data": {"brenda_reference_id": "713515", "brenda_reference_metadata": {"journal": "Skin Pharmacol. Physiol.", "pages": "160-168", "pubmed_id": "9287397", "title": "Activity and inhibition of 3-beta-hydroxysteroid dehydrogenase/delta-5-4-isomerase in human skin", "volume": "10", "year": "1997"}, "commentary": "37°C, pH not specified in the publication", "ecNumber": "1.1.1.51", "kmValue": "0.00001", "ligandStructureId": 239, "literature": [713515], "organism": "Homo sapiens", "sequence_enrichment": {"candidate_match_count": 4, "candidate_matches": [{"gene_names": ["AKR1C4"], "organism": "Homo sapiens", "protein_name": "Aldo-keto reductase family 1 member C4", "ranking_reasons": [], "ranking_score": 0.0, "sequence_length": 323, "uniprot_id": "P17516"}, {"gene_names": ["AKR1C2"], "organism": "Homo sapiens", "protein_name": "Aldo-keto reductase family 1 member C2", "ranking_reasons": [], "ranking_score": 0.0, "sequence_length": 323, "uniprot_id": "P52895"}, {"gene_names": ["AKR1C1"], "organism": "Homo sapiens", "protein_name": "Aldo-keto reductase family 1 member C1", "ranking_reasons": [], "ranking_score": 0.0, "sequence_length": 323, "uniprot_id": "Q04828"}, {"gene_names": ["HSD17B1"], "organism": "Homo sapiens", "protein_name": "17-beta-hydroxysteroid dehydrogenase type 1", "ranking_reasons": [], "ranking_score": 0.0, "sequence_length": 328, "uniprot_id": "P14061"}]}, "note": "No unique UniProt sequence match found for EC+organism lookup.", "query_organism": "Homo sapiens", "sequence_source": "unresolved_ec_organism", "sequence_source_confidence": 0.0}, "smiles_enrichment": {"matched_name": "dehydroepiandrosterone", "original_substrate": "dehydroepiandrosterone", "smiles": "CC12CCC3C(C1CCC2=O)CC=C4C3(CCC(C4)O)C", "substrate": "dehydroepiandrosterone", "smiles": "CC12CCC3C(C1CCC2=O)CC=C4C3(CCC(C4)O)C", "source_db": "brenda", "substrate": "dehydroepiandrosterone", "substrate_name_candidates": ["dehydroepiandrosterone"]}]}, "evidence_schema_version": 1, "paper_grounding_note": "Verification notes preserve a cited paper excerpt from manual review.", "paper_grounding_source": "review_notes_excerpt", "paper_grounding_status": "review_excerpt_preserved", "paper_mentions": ["37°C, pH not specified in the publication"], "paper_structure_signals": [], "sequence_resolved": true, "source_databases_merged": "brenda", "source_record_count": 1, "supporting_dois": ["10.1159/000211481"], "supporting_pmids": ["9287397", "1401999"]} </pre>
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Example 5 | data/verified_kinhub.jsonl line 4337

correction_type	PMID and sequence correction: mismatched citation -> PMID 8550415 and Q59468.
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top-level fields

source_db	brenda
ec_number	1.1.1.34
organism	Haloferax volcanii
enzyme_name	hydroxymethylglutaryl-CoA reductase (NADPH)
kcat_unit	s ⁻¹
km	0.03
km_unit	mM
kcat_over_km_unit	s ⁻¹ *mM ⁻¹
supporting_pmids	["8550415"]
supporting_dois	["10.1128/jb.178.1.19-23.1996"]
verification_status	corrected
verification_method	paper_sequence_manual_review
evidence_confidence_tier	paper_grounded
paper_grounding_status	review_excerpt_preserved

correction/evidence fields

review_summary	Corrected the mismatched PMID and resolved the enzyme sequence to UniProtKB Q59468 (hmgA) for Haloferax volcanii HMG-CoA reductase.
critique_rationale	The staged evidence contains kinetic notes that match PMID 8550415 and contradict PMID 286578, so the literature identifier required correction before sequence resolution. Among available UniProt candidates for EC 1.1.1.34 in H. volcanii, Q59468 is reviewed and gene-annotated (hmgA), making it the safest accession to anchor the sequence identity for this paper-linked batch.
agentic_understanding	The batch PMID (286578) resolves to an unrelated dentistry paper in PubMed and cannot support the Haloferax volcanii HMG-CoA reductase kinetics. PMID 8550415 (Bischoff & Rodwell, 1996) matches the exact substrate set and Km values described in the staged evidence (CoA, NADP+/NADPH, (S)-HMG-CoA, and (R,S)-mevaldehyde reactions). The enzyme sequence is therefore resolved to the reviewed UniProtKB entry Q59468 (gene hmgA), which matches the organism/strain context and provides an accession-backed amino-acid sequence.
verification_notes	Context: EC=1.1.1.34; organism=Haloferax volcanii; substrate=CoA. PMID: 8550415. DOI: 10.1128/jb.178.1.19-23.1996. Quote: "... oxidative acylation of mevaldehyde... Kms... 300 microM (CoA)..." Evidence: The abstract's oxidative acylation kinetics list CoA as a required substrate with Km ~300 microM, matching the CoA-linked batch context. Decision: Treat PMID 286578 as a mismatched citation and anchor sequence to UniProtKB Q59468 (hmgA).
audit_history	["Loop 1: Checked the staged PMID and paper clues against PubMed; PMID 286578 is unrelated to archaeal HMG-CoA reductase kinetics.", "Loop 2: Located the matching Haloferax volcanii HMGR paper (PMID 8550415) whose abstract reproduces the substrate set and Km values present in the batch evidence.", "Loop 3: Resolved sequence identity to UniProtKB Q59468 (hmgA) and recorded a paper-backed excerpt supporting the kinetic context."]

raw_data field

raw_data	<pre> {"anomaly_reasons": ["sequence_unresolved"], "evidence_confidence_note": "This record preserves review-authored paper evidence excerpts with explicit citation context.", "evidence_confidence_tier": "paper_grounded", "evidence_envelope": {"anomaly": {"reasons": ["sequence_unresolved"], "source_record_count": 1}, "identity": {"ec_number": "1.1.1.1.34", "mutation_evidence_notes": [], "mutation_evidence_sources": [], "mutation_key": "unknown_state", "mutation_signatures": [], "organisms": ["Haloferax volcanii"], "sequence_resolved": false, "sequence_sources": [], "smiles": "CC(C)(COP(=O)(O)OP(=O)(O)OCC1C(C(C(O1)N2C=NC3=C(N=CN=C32)N)O)OP(=O)(O)O)C(C(=O)NCCCC(=O)NCCCS)O", "smiles_values": ["CC(C)(COP(=O)(O)OP(=O)(O)OCC1C(C(C(O1)N2C=NC3=C(N=CN=C32)N)O)OP(=O)(O)O)C(C(=O)NCCCC(=O)NCCCS)O"], "substrate_name_candidates": ["CoA"], "uniprot_ids": []}, "identity_hints": {"cross_source_identity_matches": {"shared_organisms": ["Haloferax volcanii"], "shared_smiles_values": ["CC(C)(COP(=O)(O)OP(=O)(O)OCC1C(C(C(O1)N2C=NC3=C(N=CN=C32)N)O)OP(=O)(O)O)C(C(=O)NCCCC(=O)NCCCS)O"], "shared_substrate_name_candidates": ["CoA"], "shared_uniprot_ids": [], "sources_with_smiles": ["brenda"], "sources_with_uniprot_id": []}, "source_identity_hints": [{"organism": "Haloferax volcanii", "pubmed_id": "286578", "smiles": "CC(C)(COP(=O)(O)OP(=O)(O)OCC1C(C(C(O1)N2C=NC3=C(N=CN=C32)N)O)OP(=O)(O)O)C(C(=O)NCCCC(=O)NCCCS)O", "source_db": "brenda", "substrate_name_candidates": ["CoA"]}]}, "provenance": {"paper_mentions": [], "source_databases_merged": ["brenda"], "supporting_dois": [], "supporting_pmids": ["286578"]}, "schema_version": 1, "source_records": [{"km_unit": "mM", "km_value": 0.03, "organism": "Haloferax volcanii", "pubmed_id": "286578", "raw_data": {"ecNumber": "1.1.1.1.34", "kmValue": "0.03", "ligandStructureId": 10, "literature": [286578], "organism": "Haloferax volcanii", "sequence_enrichment": {"note": "No unique UniProt sequence match found for EC+organism lookup.", "sequence_source": "unresolved_ec_organism", "sequence_source_confidence": 0.0}, "smiles_enrichment": {"matched_name": "CoA", "original_substrate": "CoA", "smiles": "CC(C)(COP(=O)(O)OP(=O)(O)OCC1C(C(C(O1)N2C=NC3=C(N=CN=C32)N)O)OP(=O)(O)O)C(C(=O)NCCCC(=O)NCCCS)O", "substrate": "CoA", "smiles": "CC(C)(COP(=O)(O)OP(=O)(O)OCC1C(C(C(O1)N2C=NC3=C(N=CN=C32)N)O)OP(=O)(O)O)C(C(=O)NCCCC(=O)NCCCS)O", "source_db": "brenda", "substrate": "CoA", "substrate_name_candidates": ["CoA"]}]}, "evidence_schema_version": 1, "paper_grounding_note": "Verification notes preserve a cited paper excerpt from manual review.", "paper_grounding_source": "review_notes_excerpt", "paper_grounding_status": "review_excerpt_preserved", "paper_mentions": [], "sequence_resolved": false, "source_databases_merged": ["brenda"], "source_record_count": 1, "supporting_dois": ["10.1128/jb.178.1.19-23.1996"], "supporting_pmids": ["8550415"]} </pre>
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